Ezgi **Özkurt**

PostDoc researcher | Food, Microbiome and Health | Quadram Institute Bioscience

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Education

2020 **PhD**, Max Planck Institute for Evolutionary Biology.
2015 **MSc**, Middle East Technical University, Biological Sciences.
2014 **BSc**, Middle East Technical University, Molecular Biology and Genetics.

Experience

still — **PostDoc, QUADRAM INSTITUTE BIOSCIENCE, NORWICH, UK,**

- Design, develop, and implement wet lab protocols for microbiome sample processing.
- Develop and optimize pipelines for microbiome data analysis.
- Conduct meta-analyses of publicly available metagenomic datasets.
- Design, develop, and author protocols for human cohort studies.
- Former lead of the PEARL-AGE family cohort study.
- Supervise and mentor PhD, MSc, and BSc students.

Microbial enrichment methods | HMW microbial DNA extraction | Preparing sequencing libraries | Illumina sequencing | PacBio sequencing | Handling human samples | Human cohort studies | Good clinical practice | Metagenome data analysis | Microbial networks | Meta-analysis | Numerical ecology | Software development | Supervising students | Organising and training in workshops

January 2020 – January 2016 — **PhD, Max Planck Institute for Evolutionary Biology, Plön, Germany,**

Domestication-driven metaorganism evolution in wheat.

- Develop methods for microbiome extraction from low-biomass samples.
- Establish axenic host systems to study microbiome assembly.
- Perform computational analysis of microbiome sequencing datasets.
- Prepare and maintaining bacterial and fungal culture collections.

Axenic plant experiments | Low microbial biomass samples | Illumina sequencing | Microbial culture collections | Microbiome data analysis | Population Genetics | Bash scripting | R scripting

September 2015 – July 2015 — **Summer Internship, Erasmus fellowship, Max Planck Institute for Evolutionary Biology, Plön, Germany,**

Gene expression in Cottus fish in response to water temperature.

Fish experiments | Morphological analysis | ddPCR

September 2015 – February 2014 — **MSc, Biological Sciences , Middle East Technical University, ANKARA, TURKEY**

Evolution of bladder cancer investigated using exome sequencing.

Exome sequencing | R programming language | SNP analysis | Phylogenetic trees | Gene expression data analysis

October 2013 – July 2013 — **Summer Internship, Erasmus fellowship, Umea University, Umea, Sweden**

Gene silencing and Gene Tagging in Yersinia pseudotuberculosis species to understand the mechanism of Type3 secretion system.

Sterile work | PCR | Gene transformation and cloning

Awards

2023 **EMBO Scientific Exchange Grant** Received the grant as the principal applicant, allowed me to carry out my project at the Leibniz Institute on Aging (Jena, Germany)
2022 Endorsed as **Global Talent** by the Royal Society
2021 **Quadram Institute Impact Accelerator Award**, as collaborator with Dr. Falk Hildebrand, Dr. Rebecca Ansorge and Dr. Stefano Romano and Dr. Naiara Beraza : Enabling host transcriptomics on bacteria-enriched mucosal biopsies

✔ Conference Talks & Workshops

2024 Texas Medical Center, USA, Informatics Users Meeting, **Invited Speaker** : Enterosignatures : A New Perspective on the Microbiome

2024 Leibniz Institute for Aging, Germany, **Workshop Sole Organiser and Sole Trainer** : Amplicon Sequencing of Microbiomes : Theory and Hands-on Analysis

2024 University of Birmingham, UK, Data Science MSc students, **Invited Speaker** : My Journey with Microbiomes : From Plants to the Human Gut

2023 Ecology and Evolutionary Biology Symposium, Turkey, **Oral Presentation** : Conserved Gut Microbial Guilds Across the Human Life Stages

2023 Vlaams Institute for Biotechnology, Belgium, **Workshop Organiser and Trainer** : Analysis of 16S rRNA Metagenomic Experiments

2022 Ecology and Evolutionary Biology Symposium, Turkey, **Chair and Speaker** : The Establishment of Academia in Turkey & How Civil Society Organizations and Academia Collaborate and Work Together.

2022 Free University of Berlin, Germany, Colloquium Evolution and Ecology, **Invited Speaker** : My Journey with Microbiomes : From Plants to the Human Gut

2022 Microbiome Virtual International Forum, **Selected Talk** : LotuS2 : An Ultrafast and Highly Accurate Tool for Amplicon Sequencing Analysis

2022 EMBO, **Workshop Participant** : Leadership and Management Skills Course for Postdocs

2022 Earlham Institute, UK, **Workshop Organiser and Trainer** : Microbiome Community Analysis

2019 Congress of the European Society for Evolutionary Biology (ESEB), Finland, **Oral Presentation** : Higher Stochasticity of Microbiota Composition in Seedlings of Domesticated Wheat Compared to Wild Wheat.

Publications

2026 **Ezgi Özkurt**, David Schneider, Steve A James, Isabelle Hautefort, Jennifer Ahn-Jarvis, Darren Heavens, Manuel Banzhaf, Antonietta Hayhoe, Falk Hildebrand. **Enriching the Human Stool Microeukaryotes for Shotgun Sequencing** *bioRxiv; available as a pre-print*. https://doi.org/10.64898/2026.06.24.734237

2026 Alise Ponsero, Basak Bahcivanci, Antonietta Hayhoe, Animesh Acharjee, **Ezgi Özkurt**. **The Human Gut Microbiome Across the Life Course.** *FEBS Letters*. doi : https://doi.org/10.1002/1873-3468.70361

2026 Michaela Matthews, Gaëtan Thilliez, Petros Muchesa, Nicholas M. Thomson, Fiona Jena, John Macharaga, George M. Savva, Dave J. Baker, Rebecca Ansorge, **Ezgi Özkurt**, Ross Low, Neil Hall, Falk Hildebrand, Gladys Manyembere, Agnes Juru, Tapfumanei Mashe, Robert A. Kingsley. **Detection of Salmonella Typhi, associated microbial communities and AMR genes within wastewater in Harare, Zimbabwe** *VeriXiv; available as a pre-print, under review in Microbial Genomics*. https://doi.org/10.64898/2026.06.24.734237

2026 Steve James, **Ezgi Özkurt**. **Mycobiomics.** *Book chapter : Best Practice in Microbiome Research, Springer Nature Protocols Handbook*

2025 Dominic Denk, Mallika Ramakrishnan, Claire Conche, Charles Pallangyo, Marina Pesic, Kilian Kennel, Asude Kirisozu, Esther Engel, Kathleen Mohs, Birgit Ritter, Angeles Macias Pardo, **Ezgi Özkurt**, Falk Hildebrand, Melek Canan Arkan, Ari Waisman, Florian R. Greten. **IL-17RA signaling provides dual tumor suppressor function.** *Immunity*. https://doi.org/10.1016/j.immuni.2025.02.005

2025 **Ezgi Özkurt**. **Experimental Systems Are Essential for Strengthening Ecological Modeling of Microbiomes Beyond Observational Data.** *mSystems*. https://doi.org/10.1128/msystems.01745-24

2024 **Ezgi Özkurt**, Zaywa Mariush, Doreen Landermann-Habetha, Eva Stukenbrock. **Ecological assembly dynamics of the seed-borne microbiome in cultivated and wild wheat.** *Available in bioRxiv*. doi : https://doi.org/10.1101/2024.12.27.630520

2023 Santanu Das*, **Ezgi Özkurt***, Tulsi Kumari Joishy, Dibyayan Deb, Ashis K. Mukherjee, Falk Hildebrand, Mojibur R. Khan. **A single dietary factor, daily consumption of a fermented beverage, can modulate the gut microbiome within the same ethnic community.** *mSystems, *Equal contributions*. doi : https://doi.org/10.1128/msystems.00745-23

2023 Clémence Frioux, Rebecca Ansorge, **Ezgi Özkurt**, Chabname Ghassemi Nedjad, Joachim Fritscher, Christopher Quince, Sebastian M. Waszak, Falk Hildebrand. **Enterosignatures define common bacterial guilds in the human gut microbiome.** *Cell Host and Microbe*. doi : 10.1016/j.chom.2023.05.024

2023 Ali Hakimzadeh, Alejandro Abdala Asbun, Davide Albanese, Maria Bernard, Dominik Buchner, Benjamin Callahan, Gregory Caporaso, Emily Curd, Christophe Djemiel, Mikael Brandström Durling, Vasco Elbrecht, Zachary Gold, Hyun Gweon, Mehrdad Hajibabaei, Falk Hildebrand, Vladimir Mikryukov, Eric Normandeau, **Ezgi Özkurt**, Jonathan M Palmer, Géraldine Pascal, Teresita Porter, Daniel Straub, Martti Vasar, Tomáš Větrovský, Haris Zafeiropoulos, Sten Anslan. **A pile of pipelines : an overview of the bioinformatics software for metabarcoding data analyses.** *Molecular Ecology Resources*. https://doi.org/10.1111/1755-0998.13847

2022 **Ezgi Özkurt**, Joachim Fritscher, Nicola Soranzo, Duncan Y. K. Ng, Robert P. Davey, Mohammad Bahram, Falk Hildebrand. **LotuS2 : An ultrafast and highly accurate tool for amplicon sequencing analysis.** *Microbiome*. https://doi.org/10.1186/s40168-022-01365-1

2021 Falk Hildebrand, Toni I Gossmann, Clémence Frioux, **Ezgi Özkurt**, Pernille Neve Myers, Pamela Ferretti, Michael Kuhn, Mohammad Bahram, Henrik Bjørn Nielsen, Peer Bork. **Dispersal strategies shape persistence and evolution of human gut bacteria.** *Cell Host and Microbe*. doi : 10.1016/j.chom.2021.05.008

2021 **Ezgi Özkurt**, Falk Hildebrand. **Lifelong sex-dependent trajectories of the human gut microbiota.** *Nature Ageing*. https://doi.org/10.1038/s43587-020-00019-x

2021 Janis R Bedarf, Naiara Beraza, Hassan Khazneh, **Ezgi Özkurt**, David Baker, Valeri Borger, Ullrich Wüllner, Falk Hildebrand. **Much ado about nothing? Off-target amplification can lead to false-positive bacterial brain microbiome detection in healthy and Parkinson's disease individuals.** *Microbiome*. https://doi.org/10.1186/s40168-021-01012-1

2021 Etka Yapar, Ekin Saglican, Handan Melike Dönertaş, **Ezgi Özkurt**, Zheng Yan, Haiyang Hu, Song Guo, Babür Erdem, Rori V Rohlfs, Philipp Khaitovich, Mehmet Somel. **Convergent evolution of primate testis transcriptomes reflects mating strategy.** *Available as a preprint in bioRxiv*. doi : https://doi.org/10.1101/010553

2021 Ryszard Soluch, Nils F Hülter, Devani Romero Picazo, **Ezgi Özkurt**, Eva H. Stukenbrock, Tal Dagan. **Colonization dynamics of Pantoea agglomerans in the wheat root habitat.** *Environmental Microbiology*. https://doi.org/10.1111/1462-2920.15430

2020 **Ezgi Özkurt**, M. Amine Hassani, Ugur Sesiz, Tal Dagan, Hakan Özkan, Eva H. Stukenbrock. **Seed-derived microbial colonization of wild emmer and domesticated bread wheat (Triticum dicoccoides and T. aestivum) seedlings shows pronounced differences in overall diversity and composition.** *mBio*. https://doi.org/10.1128/mbio.02637-20

2019 M. Amine Hassani, **Ezgi Özkurt**, S. Franzenburg, Eva H. Stukenbrock. **Ecological assembly processes of the bacterial and fungal microbiota of wild and domesticated wheat.** *Phytobiomes*. https://doi.org/10.1094/PBIOMES-01-20-0001-SC